Pharmaceutical Biotechnology

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

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Pharmaceutical Biotechnology (A32402)

Short Title: Pharmaceutical Biotechnology
Faculty Science and Computing
Department Science
Cluster Biology
Author CLENNON
Credits 5 **Level:** Intermediate

Description of Module / Aims

This module will describe the application of biotechnology to the production of pharmaceutical products. Topics covered will include antibiotics, antibody-based drugs and proteins. The use and advantages of biotransformations and biocatalysis for the production of bio-organic products will be presented. Selectivity of such reactions and its importance is highlighted.

Programmes

BIOT-0001	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	3 / 5 / M
BIOT-0001	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	3 / 5 / M
BIOT-0001	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	3 / 5 / M
BIOT-0001	BSc in Pharmaceutical Science (WD_SPHSC_D)	3 / 5 / M

Indicative Content

- Detailed treatment of design issues around bioreactors; sterilisation, including calculation of D & Z values; sterile delivery and sampling; clean room technology
- Formulation of biotechnology products, including biopharmaceutical considerations of sterility and stability
- Downstream processing for product isolation and purification
- Biological approaches to drug design and delivery; cultivation systems; biological pharmaceutical products such as insulin and interferon; protein purification
- Production of biopharmaceuticals such as antibiotics and antibody-based drugs; recombinant DNA technology; expression systems; immunotherapeutic agents and immunotechnology
- Ethical considerations in animal testing and marketing
- Enzymes, background, classification, kinetics of enzymatic reactions, mechanism of action
- Introduction to stereoselective synthesis of pharmaceuticals, resolution, asymmetric synthesis from prochiral starting materials
- Biocatalysis for the synthesis of organic biomolecules; use of microbial and enzymatic catalysts and their advantages in terms of chemoselectivity, regioselectivity and stereoselectivity
- Important enzymes for such biotransformations, including hydrolases and oxidoreductases; enzymes in organic solvents; immobilised cells and enzymes
- Screening techniques for the identification of enzymes and microbial isolates with biocatalytic potential
- Analytical methods for the estimation of enantiomeric excess, including optical methods, chiral chromatography and polarimetry
- Practical work related to the above topics, screening microbes for potential biopharmaceutical compounds, protein purification methods, fermentations and enzymes in organic solvents for synthesis of biomolecules, gas chromatography (GC) and polarimetry for product analysis.; analytical methods for biopharmaceuticals

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Determine the role of biotechnology in the production of biopharmaceutical products.
2. Examine the main features of biotechnology, including processing and the critical control parameters in production.
3. Analyse the main groups of biologically-active products with pharmaceutical applications, including antibiotics, antibody-based drugs and therapeutic proteins.
4. Explore the role of the biocatalysts in the synthesis of organic drugs with the application of specific enzymes and their reactions.
5. Determine the advantages of using biotransformations in the production of such compounds and detail how in particular stereoselectivity may be measured analytically.
6. Demonstrate proficiency in laboratory techniques related to the above outcomes.

Learning and Teaching Methods

- Lectures.
- Laboratory-based practicals.
- Self-directed study.
- Tour to relevant facility.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Practical	50%	6
In-Class Assessment	30%	1,2,3,4,5
Assignment	20%	1,2,3,4,5

Assessment Criteria

<40%: Little or no demonstration of an understanding of the fundamentals of pharmaceutical biotechnology. Lack of competence with associated laboratory skills and practices.

40%-49%: Ability to show a limited understanding of the fundamentals of pharmaceutical biotechnology. Will display some ability with associated laboratory skills and practices.

50%-59%: As well as the above, ability to show an understanding of some of the complexities of pharmaceutical biotechnology. Will display reasonable competence with associated laboratory skills and practices.

60%-69%: In addition, the student will demonstrate a more detailed knowledge of pharmaceutical biotechnology and display very good competence with associated laboratory skills and practices.

70%-100%: All of the above to an excellent level. In addition, ability to demonstrate a comprehensive knowledge and understanding of pharmaceutical biotechnology and show ability to evaluate and relate topics. Will also demonstrate excellent competence and independence with associated laboratory skills and practices.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- "Web-based searching for review papers on the topic." http://library.wit.ie/
- Campbell, N.A., J.B. Reece, L.A. Urry, M.L. Cain, S.A. Wasserman and Minorsky, P.V., Jackson, R.B. . _Biology_. 8th ed. San Franscisco: Pearson, 2000.
- Crommelin, D.J., R.D. Sindelar and B. Meibohm. _Pharmaceutical Biotechnology_. 4th ed. New York: Springer-Verlag New York, 2013.

Supplementary Material

- Junhua , T. and R. Kazlauskas. _Biocatalysis for green chemistry and chemical process development_. Hoboken: John Wiley and Sons, 2011.
- Liese, A. _Industrial Biotransformations - A Collection of Processes_. Weinheim, Germany: Wiley-VCH, 2000.
- Ramesh, N.P. _Biocatalysis in the pharmaceutical and biotechnology industries_. 2nd ed. London: CRC Press, 2006.
- Roberts, S.M. _Biocatalysts For Fine Chemical Synthesis_. England: Wiley, 1999.
- Tao, J., A. Leise and G.Q. Lin. _Biocatalysis for the Pharmaceutical Industry : Discovery, Development and Manufacturing_. Hoboken: John Wiley and Sons, 2009.

Requested Resources

- Room Type: Chemistry Lab
- Room Type: Biology Lab
- Lecture Room: Loose Seated